

Assignment 03: Working with Arrays (Lists) in Python

Instructions

1. Write each program in a separate Python file.
 2. Use meaningful variable names and proper indentation.
 3. Display clear output messages.
 4. Use loops wherever required.
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1 Find the Average of Elements in an Array

Problem: Write a Python program to calculate the **average** of all numbers stored in a list.

Example Output:

```
Numbers = [5, 10, 15, 20, 25]
Average = 15.0
```

2 Count Positive, Negative, and Zero Values

Problem: Create a list of integers and count how many numbers are **positive**, **negative**, and **zeros**.

Example Output:

```
Numbers = [5, -2, 0, 8, -7, 0, 3]
Positive = 3
Negative = 2
Zeros = 2
```

3 Copy Elements from One Array to Another

Problem: Write a program to copy all elements of one list into another list using a loop.

Example Output:

```
Original List = [10, 20, 30, 40]
Copied List = [10, 20, 30, 40]
```

4 Find the Second Largest Element

Problem: Write a Python program to find the **second largest** element in a list.

Example Output:

```
Numbers = [12, 45, 7, 34, 89, 23]  
Second Largest = 45
```

 5 Separate Even and Odd Numbers into Two Lists

Problem: Create a list of integers and separate them into **two new lists** — one for even numbers and another for odd numbers.

Example Output:

```
Numbers = [11, 22, 33, 44, 55, 66]  
Even = [22, 44, 66]  
Odd = [11, 33, 55]
```